

ROHIT R VERNEKAR

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PROFILE

Computer Science undergraduate (BE, GPA 8.4) specialising in backend engineering with hands-on experience building production-grade systems in Java and Python. Published researcher, hackathon winner, and independent builder spanning adversarial ML security, REST API design, and web privacy tooling. Seeking backend or full-stack engineering roles on high-impact teams.

PROFESSIONAL EXPERIENCE

Java Development Intern — Oasis Infobyte (AICTE OIB-SIP)

Feb – Mar 2025

- Developed **4 production-quality Java console applications** (including a fully functional ATM interface and multi-mode game) by effectively applying OOP design patterns, encapsulation, and core data structures within the AICTE-recognized framework.

CORE SKILLS

- Java • Python • Spring Boot • Django • FastAPI • Flask • REST APIs • Adversarial ML • XAI
- GitHub • Postman • SQL

PROJECTS

NEURALARMOR — ADVERSARIAL AI SECURITY MIDDLEWARE

Major Project

- Engineered a model-agnostic security reverse proxy in FastAPI to protect autonomous perception systems; achieved a **98% reduction in inference latency** (5s to 0.1s) using a CVPR 2025 Adversarial Consistency Distillation purifier.
- Built a Red Teaming scanner using Adversarial Robustness Toolbox (ART) to stress-test models against PGD/FGSM attacks; integrated Grad-CAM XAI heatmaps for forensic noise visualisation.

DIGITAL FOOTPRINT ANALYZER — REAL-TIME WEB PRIVACY RISK AUDITOR

Academic Mini Project

- Built a privacy toolkit (**mitmproxy**, **Flask**, Manifest V3 Chrome extension) that tracks fingerprinting signals and cookies to compute real-time **0–100 domain risk scores**, entirely via non-intrusive network analysis and PostgreSQL data visualization.

CONNECTNET — OFFLINE-FIRST EMERGENCY RESCUE PLATFORM

VIDYUT 2K25 Hackathon • 2nd Place

- Developed an offline-first Android SOS platform utilizing **BLE/Wi-Fi Direct** mesh networking, Room DB, and Firebase to sync and visualize live emergency alerts on a Google Maps dashboard—built entirely during a **6-hr hackathon sprint**.

EDUCATION & CERTIFICATION

B.E in Computer Science & Design — BGS College of Engineering and Technology

Expected Jun 2027 | GPA 8.4

- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate — Oracle University

RESEARCH PUBLICATION

BOOK CHAPTER — IIP PROCEEDINGS, VOL 6, BOOK 2 • "MULTI-VECTOR GUARDRAILS"

Peer-Reviewed Author

- Authored an approved book chapter on Knowledge Engineering for autonomous systems. Proposed a dual-path processing architecture (Probabilistic AI vs. Deterministic Backend) to guarantee ethical alignment, grounded in empirical data from the NeuralArmor system.

KEY ACHIEVEMENTS

- Undergraduate Researcher:** One of few undergraduates to clear peer review for an original book chapter in IIP Proceedings.
- Hackathon Silver Medalist:** Won **2nd place** at the IEEE Bangalore Section-affiliated VIDYUT 2K25 hackathon.